Fiscal Year:	FY 2025	**Task Last Updated:**	FY 04/21/2025
PI Name:	Santa Maria, Sergio Ph.D.		
Project Title:	Acquisition of Beneficial Mutations Through Adaptive Evolution Under Simulated Microgravity		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground,New Investigation	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	06/01/2021	**End Date:**	12/31/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2024 per F. Hernandez/ARC (Ed., 10/27/23) NOTE: End date changed to 12/31/2023 per F. Hernandez/ARC (Ed., 6/2/23) NOTE: End date changed to 02/28/2023 per F. Hernandez/ARC (Ed., 4/25/22)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sherlock, Gavin Ph.D. (Stanford University)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Adaptive evolution plays a major role in the appearance of new phenotypes in nature and is of key importance in biomedical research, including the onset of carcinogenesis and in the emergence of drug-resistant microorganisms. The harsh environment of space provides the ideal environment for the acquisition of adaptive mutations, particularly in microbial organisms. Experimental evolution studies in the laboratory will allow us to better understand how specific mutations arise under spaceflight conditions. Our primary goal is to investigate how a eukaryotic organism responds to simulated microgravity after multiple generations. S. cerevisiae or budding yeast is a single-cell model eukaryote that has been widely used in biomedical research, primarily due to its well-characterized biology, easy manipulation, short division time, and, most importantly, its similarities with human cell functions. We hypothesize that beneficial mutations acquired through multiple generations under simulated microgravity will render cells with increased growth and/or metabolic rates, and phenotypes different to those observed in 1g controls. Aim1 involves the identification of adaptive beneficial mutations via experimental evolution and next-generation sequencing. Barcoded yeast libraries will be grown using rotating wall vessels (RWVs) for multiple generations. DNA barcodes are unique 20-nucleotide sequences that are used to “tag” or “barcode” individual strains and allow us to follow the appearance of these adaptive mutations and to identify which mutations are actually beneficial to survive under simulated microgravity. The frequency of the resulting yeast lineages and the appearance of adaptive mutations can be then followed by genomic DNA extraction followed by amplification and sequencing. Aim 2 involves the generation of mutational spectra from the surviving yeast at different time points during multigenerational growth. We propose to perform mutation analyses via whole genome sequencing to generate mutation spectra after long-term exposure to simulated microgravity. The resulting acquired mutations will be then compared to those of the non-exposed 1g controls and to previous generation spectra. All RWV growth studies will be performed at Ames Research Center. Barcoded yeast libraries developed by the Sherlock team at Stanford University are already available. DNA amplification and next-generation sequencing will be performed at Stanford. All project data will be archived in the NASA GeneLab repository. Studies like the one proposed here will allow us to identify beneficial mutations responsible for an adaptive response to space-like conditions. The proposed work will address questions on how spaceflight affects microbial growth and population dynamics (Space Biology Microbiology objectives A-C), and how it alters DNA function and intracellular processes (Cell and Molecular Biology objectives B and E). We will try to identify what processes are influenced by the spaceflight environment, and whether long-term exposure has biological effects across multiple generations (Guiding Questions MB-1, MB-2, MB-5, MB-6, CMB- 1, CMB-7, and Decadal Survey recommendations P1, P2, CC9, and AH16, https://).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The significance of exploring the effects of the space environment on microbes (and their adaptation) has important implications. For example, abnormal cell growth and physiology – especially in pathogens – could be detrimental to astronaut health, especially considering NASA’s upcoming long-term missions to outer space. On the other hand, altered microbial growth in space could be advantageous for the production of high-value products, including medicines, vitamins, and food.
Task Progress:	Aim 1: To investigate the acquisition of adaptive beneficial mutations via experimental evolution and high-throughput sequencing. After optimization of the growth conditions, we performed a 21-day evolution experiment using rotating wall vessels (RWVs). Samples from four RWVs under simulated microgravity and one RWV for the 1g control (5 RWVs total per time point) were collected every 24 hours (i.e., at every ~8 generations). The evolution experiments were completed without issues and the samples were transferred to the team at Stanford University, where yeast samples underwent DNA extraction and amplification. After sample extraction, lineage tracking was performed by sequencing of the DNA barcodes from the the 0, 40, 80, 120, and 168 generations to determine whether there was expansion of any of the barcodes, indicating that adaptation had taken place. Interestingly, we observed that there was substantial adaptation in all the experimental samples and the 1g control. Aim 2: To explore the mutational spectrum from the surviving yeast at different time points during multigenerational growth. Whole genome, whole population sequencing was performed to determine whether there were beneficial mutations that had arisen to a high enough frequency for us to detect. Many mutations in genes known to be involved in determining the growth rate were observed (IRA1 and IRA2 genes). Such mutations were present in both the experimental and control samples, suggesting that at least one selective pressure they were experiencing was to simply grow faster in the growth medium. We also observed mutations in the GAT1 gene, which is involved in nitrogen catabolism – such mutations were at higher frequency in the experimental vessels compared to the control. Finally, we observed many mutations in the LST8 gene, whose encoded protein is known to be involved in the regulation of cell growth in response to nutrient availability and cellular stresses. To determine whether any of these mutations provided a fitness benefit specific to the experimental condition (i.e., simulated microgravity), we performed additional fitness tests by combining our clones with others known to be adaptive in prior glucose and nitrogen limited evolutions. The fitness of clones in the pool was then measured in both the experimental and control conditions via serial transfer and barcode sequencing. Fitness between the conditions was compared. We found that clones were more fit in both conditions, with largely similar fitness in each. Based on these results, we conclude that, at least under our evolution conditions, that the experimental and control samples’ environments were insufficiently different to result in different evolutionary outcomes. It is unclear whether this is because microgravity cannot provide a distinct selective pressure, or if, instead, it is that the medium provides too strong of a selective pressure compared to that which might be exerted by microgravity.
Bibliography Type:	Description: (Last Updated: 12/08/2025)
Articles in Peer-reviewed Journals	Lambacher A, Akiyama K, Lingam N, Shuman D, Santa Maria SR. "LEIA: NASA’s first biological mission on the lunar surface since 1972." Acta Astronaut. 2025 Jun;231:148-52. https://doi.org/10.1016/j.actaastro.2025.02.050 , Jun-2025

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